

Solving Logarithmic Equations with Domain Checks

Answer Key

Precalculus

Quick Answers

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|---|--|
| 1. 32 | 7. 8 |
| 2. 5 | 8. $-8, 3$ |
| 3. 6 | 9. (a) domain = $(0, \infty)$, (b) $x = 2$ |
| 4. 82 | 10. 27 |
| 5. 7 | 11. (a) domain = $(8, \infty)$, (b) $x = 9$ |
| 6. (a) domain = $(3, \infty)$, (b) $x = 5$ | 12. (a) domain = $(3, \infty)$, (b) $x = 5$ |
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Curriculum

Level: Precalculus

HSF-BF.B.5 – *problems 2, 3, 5, 6, 8, 9, 11, 12*

HSF-LE.A.4 – *problems 1, 4, 7, 10*

Difficulty 1–5 of 5 across 16 tagged checks.

- 1.** Solve for x : $\log_2 x = 5$

Solution: The definition of a logarithm says $\log_b A = c$ means $A = b^c$. Here $x = 2^5 = 32$, and $32 > 0$, so the domain allows it.

$x = 32$

- 2.** Solve for x : $\log_6(4x) = \log_6 20$

Solution: Same base on both sides and the log function is one-to-one, so the arguments are equal: $4x = 20$, giving $x = 5$. Check: $4(5) = 20 > 0$.

$x = 5$

- 3.** Solve for x : $\log_3 x + \log_3 4 = \log_3 24$

Solution: Condense the left side with the product rule, then equate arguments.

$$\begin{aligned}\log_3(4x) &= \log_3 24 \\ 4x &= 24\end{aligned}$$

So $x = 6$, and $6 > 0$ so both original logarithms are defined.

$x = 6$

- 4.** Solve for x : $\log_3(x - 1) = 4$

Solution: Convert to exponential form: $x - 1 = 3^4 = 81$, so $x = 82$. Check the argument: $82 - 1 = 81 > 0$.

$x = 82$

5. Solve for x : $\log_5(2x - 3) = \log_5(x + 4)$

Solution: Equate the arguments: $2x - 3 = x + 4$, so $x = 7$. Check both arguments at $x = 7$: $2(7) - 3 = 11 > 0$ and $7 + 4 = 11 > 0$.

$$x = 7$$

6. $\log x + \log(x - 3) = 1$. (a) Domain. (b) Solve.

Solution (a): $\log x$ needs $x > 0$; $\log(x - 3)$ needs $x - 3 > 0$, i.e. $x > 3$. Both hold exactly when

$$x > 3$$

Solution (b): Condense and convert:

$$\begin{aligned}\log(x(x - 3)) &= 1 \\ x^2 - 3x &= 10 \\ x^2 - 3x - 10 &= 0 \\ (x - 5)(x + 2) &= 0\end{aligned}$$

The algebra offers $x = 5$ and $x = -2$, but -2 fails the domain from part (a), so it is extraneous.

$$x = 5$$

7. Solve for x : $\log_{1/2} x = -3$

Solution: Convert to exponential form with the fractional base:

$$x = \left(\frac{1}{2}\right)^{-3} = 2^3 = 8$$

A negative exponent on a reciprocal base flips it back. Check: $8 > 0$.

$$x = 8$$

8. Solve for x : $\log_6(x^2 + 5x) = \log_6 24$

Solution: Equate arguments, then test each candidate — do not discard the negative one on sight.

$$\begin{aligned}x^2 + 5x &= 24 \\ x^2 + 5x - 24 &= 0 \\ (x + 8)(x - 3) &= 0\end{aligned}$$

At $x = 3$: $9 + 15 = 24 > 0$ ✓. At $x = -8$: $64 - 40 = 24 > 0$ ✓. The argument here is a quadratic, not a linear expression, so a negative x can still make it positive. Both values are solutions.

$$x = -8 \quad \text{or} \quad x = 3$$

9. $\log_4(x + 6) - \log_4 x = 1$. (a) Domain. (b) Solve.

Solution (a): The arguments need $x + 6 > 0$ (so $x > -6$) and $x > 0$. The stricter of the two governs, so the domain is

$$x > 0$$

Solution (b): Condense with the quotient rule, then convert:

$$\begin{aligned}\log_4\left(\frac{x+6}{x}\right) &= 1 \\ \frac{x+6}{x} &= 4 \\ x+6 &= 4x\end{aligned}$$

So $3x = 6$ and $x = 2$, which satisfies $x > 0$.

$x = 2$

10. Solve for x : $\log_9 x = \frac{3}{2}$

Solution: Convert, then read the fractional exponent as a root followed by a power:

$$x = 9^{3/2} = \left(\sqrt{9}\right)^3 = 3^3 = 27$$

Check: $27 > 0$.

$x = 27$

11. $\log_3 x + \log_3(x-8) = 2$. **(a)** Domain. **(b)** Solve.

Solution (a): $x > 0$ and $x - 8 > 0$. The second is stricter, so the domain is

$x > 8$

Solution (b): Condense with the product rule and convert:

$$\begin{aligned}\log_3(x(x-8)) &= 2 \\ x^2 - 8x &= 3^2 = 9 \\ x^2 - 8x - 9 &= 0 \\ (x-9)(x+1) &= 0\end{aligned}$$

$x = -1$ is outside $x > 8$, so it is extraneous; $x = 9$ satisfies it.

$x = 9$

12. $\ln(x+3) + \ln(x-3) = \ln 16$. **(a)** Domain. **(b)** Solve.

Solution (a): $x+3 > 0$ gives $x > -3$; $x-3 > 0$ gives $x > 3$. The stricter one governs, so the domain is

$x > 3$

Solution (b): Condense the left side, then equate arguments:

$$\begin{aligned}\ln((x+3)(x-3)) &= \ln 16 \\ x^2 - 9 &= 16 \\ x^2 &= 25\end{aligned}$$

The square root step yields $x = 5$ and $x = -5$. Only $x = 5$ lies in $x > 3$; $x = -5$ would make both original arguments negative.

$x = 5$